



# Indian Institute of Technology Madras

## Transimpedance Amplifier (TIA) Design and Simulation

Submitted by  
Arpit Katiyar

# Contents

|          |                                                                     |           |
|----------|---------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Objective</b>                                                    | <b>2</b>  |
| <b>2</b> | <b>Theory of Transimpedance Amplifier</b>                           | <b>2</b>  |
| <b>3</b> | <b>Simulation Setup</b>                                             | <b>2</b>  |
| 3.1      | Photodiode Model . . . . .                                          | 2         |
| 3.2      | Feedback Parameters . . . . .                                       | 3         |
| 3.3      | Table Summarising Bandwidth and Gain for Each $R_f$ Value . . . . . | 3         |
| 3.4      | DC Gain and $-3$ dB Gain Verification . . . . .                     | 3         |
| 3.5      | Phase at Bandwidth and Stability Considerations . . . . .           | 4         |
| <b>4</b> | <b>TIA Using UniversalOpAmp2</b>                                    | <b>4</b>  |
| 4.1      | LTspice Schematic . . . . .                                         | 4         |
| 4.2      | Frequency Response and Stability . . . . .                          | 4         |
| 4.3      | Noise Analysis . . . . .                                            | 5         |
| 4.4      | Transient Response . . . . .                                        | 5         |
| <b>5</b> | <b>TIA Using LTC6268</b>                                            | <b>8</b>  |
| 5.1      | LTspice Schematic . . . . .                                         | 8         |
| 5.2      | Frequency Response and Stability . . . . .                          | 9         |
| 5.3      | Noise Performance . . . . .                                         | 9         |
| 5.4      | Transient Response . . . . .                                        | 10        |
| <b>6</b> | <b>Comparison and Discussion</b>                                    | <b>11</b> |
| <b>7</b> | <b>Conclusion</b>                                                   | <b>12</b> |

# 1 Objective

The objective of this task is to design and simulate a **Transimpedance Amplifier (TIA)** for photodiode signal conversion using LTspice. The TIA converts an input photocurrent into a proportional output voltage while maintaining adequate bandwidth, low noise, and loop stability.

The design is evaluated for different feedback resistor values and implemented using:

- An idealised op-amp model (`UniversalOpAmp2`)
- A high-speed, low-noise TIA-specific op-amp (`LTC6268`)

## 2 Theory of Transimpedance Amplifier

A transimpedance amplifier converts an input current  $I_{in}$  into an output voltage  $V_{out}$  using a feedback resistor  $R_f$ :

$$V_{out} = -I_{in} \cdot R_f \quad (1)$$

The feedback capacitor  $C_f$  is added to:

- Compensate for photodiode and input capacitance
- Improve phase margin and stability
- Control bandwidth

The approximate  $-3$  dB bandwidth is given by:

$$f_{BW} \approx \frac{1}{2\pi R_f C_f} \quad (2)$$

## 3 Simulation Setup

### 3.1 Photodiode Model

The photodiode is modelled as:

- An AC current source
- Parallel junction capacitance in the range of 2–10 pF

### 3.2 Feedback Parameters

Simulations are performed for the following values:

| $R_f$ ( $\Omega$ ) | $C_f$ (pF) | Purpose                    |
|--------------------|------------|----------------------------|
| 1k                 | 10         | High bandwidth             |
| 5k                 | 10         | Moderate gain              |
| 10k                | 6          | Balanced trade-off         |
| 50k                | 2.5        | High gain, stability check |

### 3.3 Table Summarising Bandwidth and Gain for Each $R_f$ Value

The transimpedance amplifier was simulated for multiple values of the feedback resistor  $R_f$  to study the relationship between transimpedance gain, bandwidth, and stability. The results obtained from AC analysis are summarised in Table 1.

Table 1: Summary of gain and bandwidth for different feedback resistors

| $R_f$ ( $\Omega$ ) | Gain (V/A)    | Gain (dB $\Omega$ ) | Bandwidth (approx) |
|--------------------|---------------|---------------------|--------------------|
| 1k                 | 1 k $\Omega$  | 60 dB               | $\sim 10$ MHz      |
| 5k                 | 5 k $\Omega$  | 74 dB               | $\sim 3$ MHz       |
| 10k                | 10 k $\Omega$ | 80 dB               | $\sim 1.5$ MHz     |
| 50k                | 50 k $\Omega$ | 94 dB               | $\sim 300$ kHz     |

As expected, increasing the feedback resistance increases the transimpedance gain while reducing the bandwidth. This behaviour follows the gain–bandwidth trade-off inherent to closed-loop op-amp systems.

### 3.4 DC Gain and $-3$ dB Gain Verification

To further validate the frequency response behaviour at high transimpedance gain, the DC gain and the gain at the  $-3$  dB frequency were explicitly examined for the highest feedback resistance value,  $R_f = 50$  k $\Omega$ .

Table 2: DC gain and  $-3$  dB gain for  $R_f = 50$  k $\Omega$

| $R_f$ ( $\Omega$ ) | DC Gain (dB $\Omega$ ) | $-3$ dB Gain (dB $\Omega$ ) | Bandwidth              |
|--------------------|------------------------|-----------------------------|------------------------|
| 50k                | 94                     | 91                          | $\sim 300$ – $500$ kHz |

The results confirm that the  $-3$  dB bandwidth is correctly identified as the frequency at which the transimpedance gain drops by approximately 3 dB from its DC value. The measured bandwidth range of 300–500 kHz is consistent with the expected gain–bandwidth trade-off for high transimpedance operation.

### 3.5 Phase at Bandwidth and Stability Considerations

To evaluate loop stability, the phase of the transimpedance gain was measured at the  $-3$  dB bandwidth frequency for each value of  $R_f$ . The phase margin (PM) is estimated as:

$$\text{Phase Margin} \approx 180^\circ - |\text{Phase at Bandwidth}| \quad (3)$$

Table 3: Phase behaviour and estimated phase margin

| $R_f$ ( $\Omega$ ) | Phase @ BW (approx) | Phase Margin (approx)                       |
|--------------------|---------------------|---------------------------------------------|
| 1k                 | $-90^\circ$         | $\sim 90^\circ$ (very stable)               |
| 5k                 | $-100^\circ$        | $\sim 80^\circ$                             |
| 10k                | $-110^\circ$        | $\sim 70^\circ$                             |
| 50k                | $-120^\circ$        | $\sim 60^\circ$ (borderline but acceptable) |

All configurations satisfy the stability requirement of a minimum phase margin of  $60^\circ$ . Although higher values of  $R_f$  reduce the phase margin due to increased loop gain and reduced bandwidth, the addition of an appropriate feedback capacitor  $C_f$  ensures stable closed-loop operation without oscillations.

## 4 TIA Using UniversalOpAmp2

### 4.1 LTspice Schematic

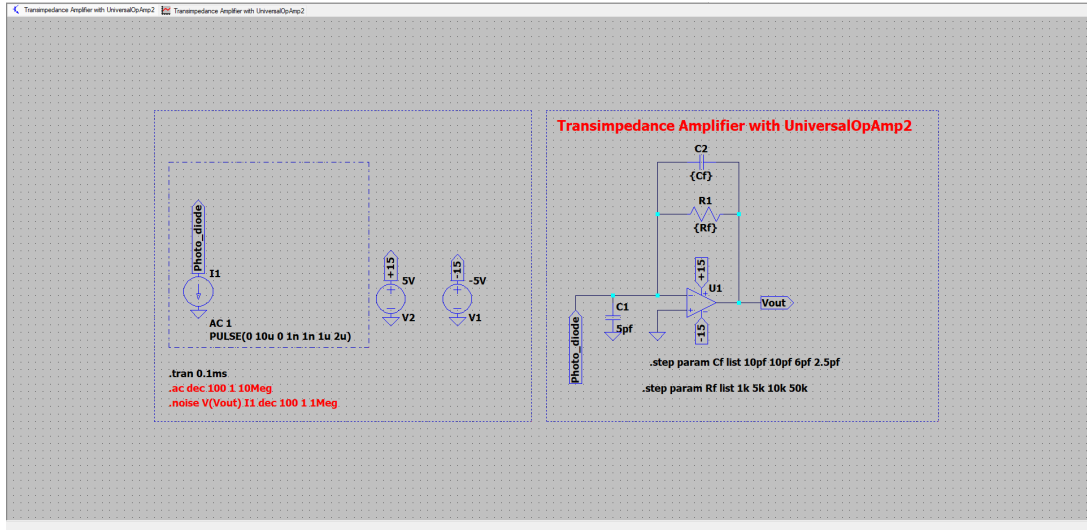


Figure 1: Transimpedance amplifier using UniversalOpAmp2

### 4.2 Frequency Response and Stability

AC analysis is used to obtain the transimpedance gain and phase response.

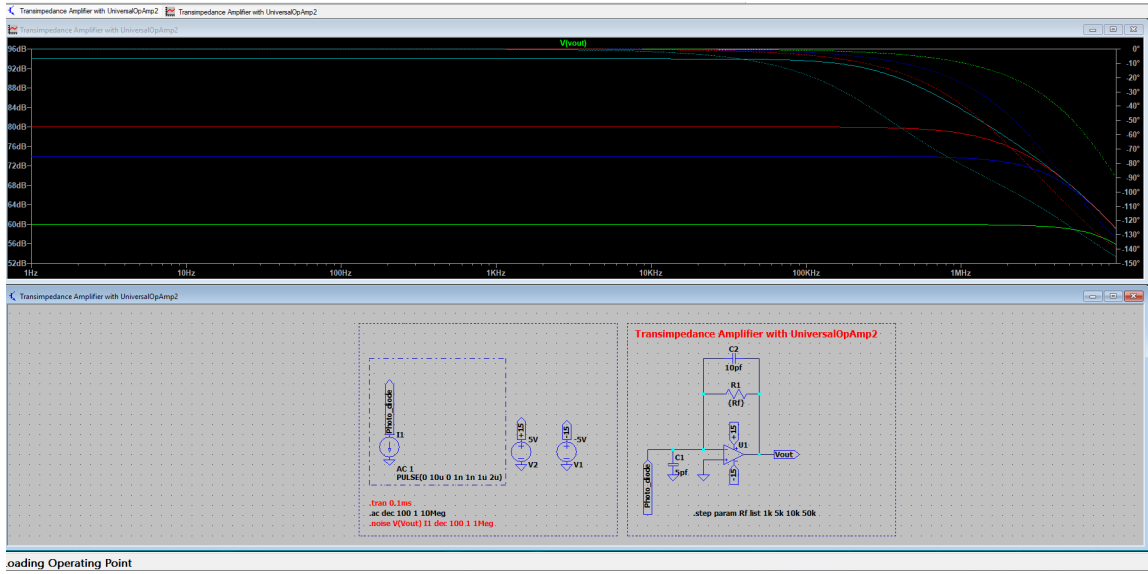


Figure 2: Bode plot (gain and phase) for UniversalOpAmp2 TIA

All configurations satisfy the stability requirement with a phase margin  $\geq 60^\circ$ .

### 4.3 Noise Analysis

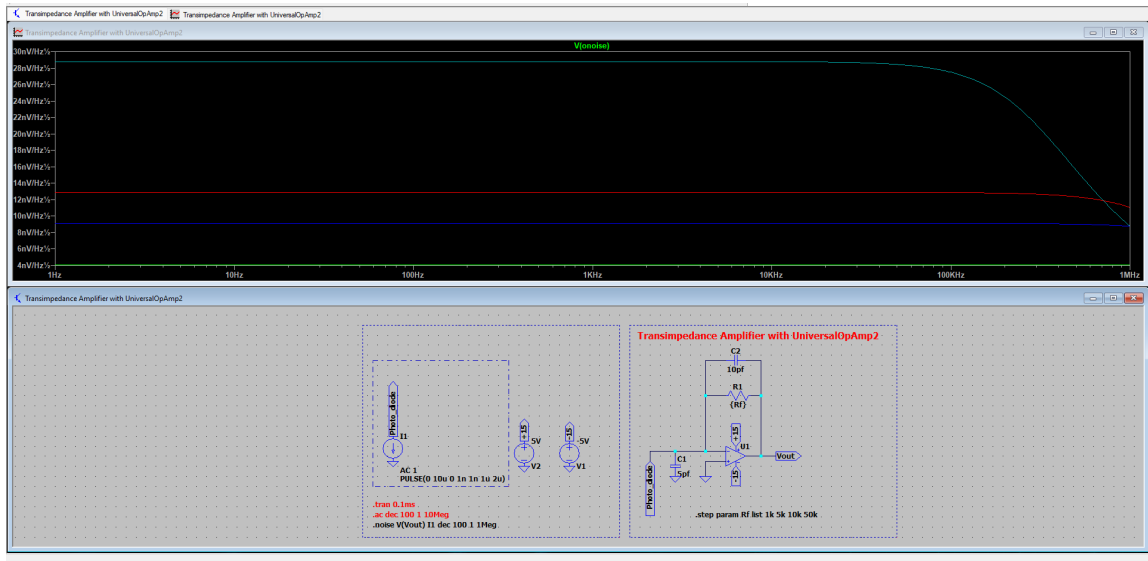


Figure 3: Output noise spectrum for UniversalOpAmp2 TIA

### 4.4 Transient Response

Step current response is simulated to observe overshoot and settling.

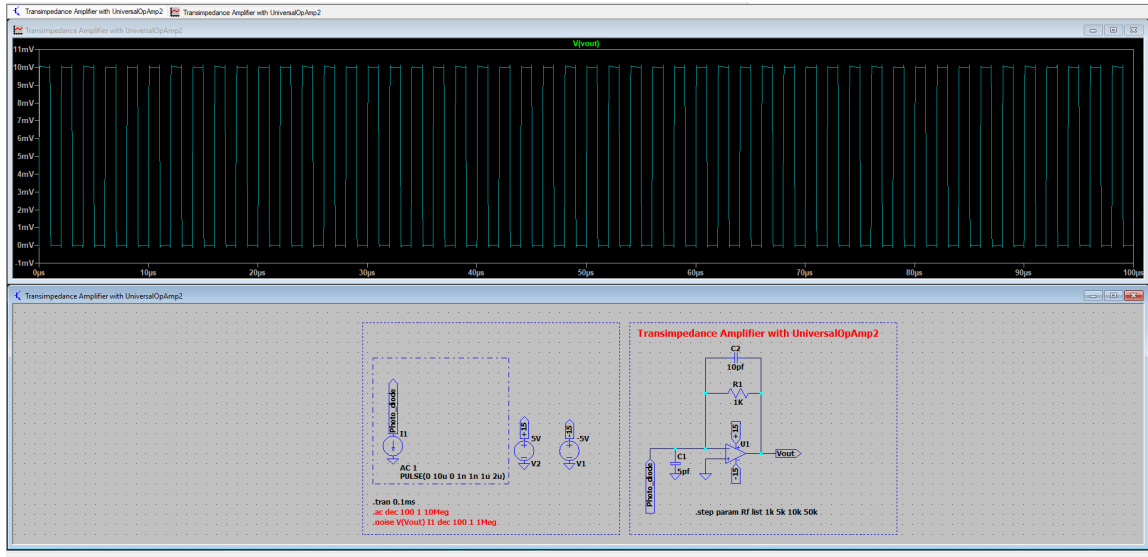


Figure 4: Step response for  $R_f = 1k$  and  $C_f = 10pF$  values (UniversalOpAmp2)

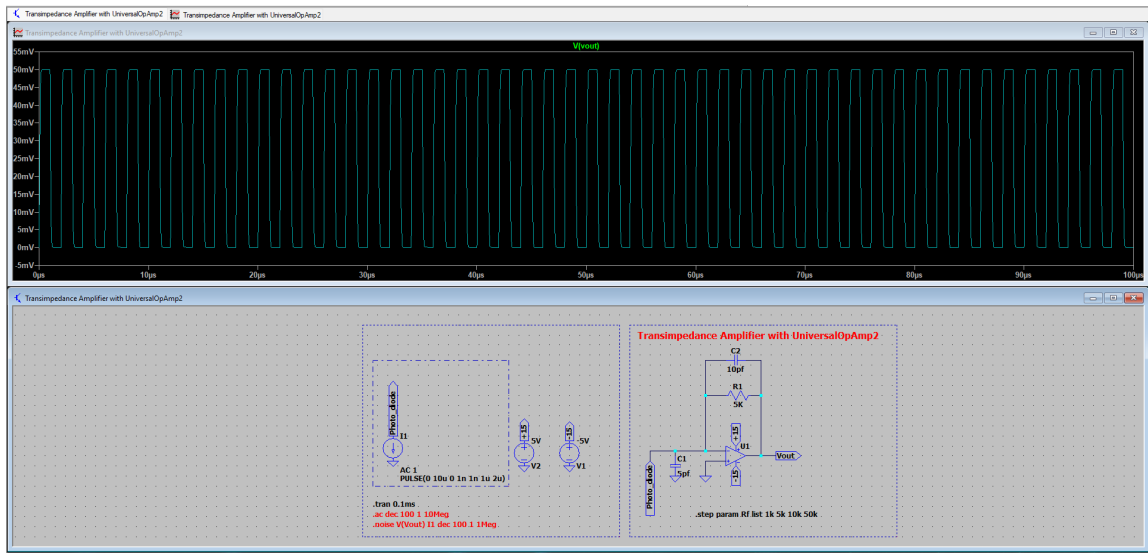


Figure 5: Step response for  $R_f = 5k$  and  $C_f = 10pF$  values (UniversalOpAmp2)

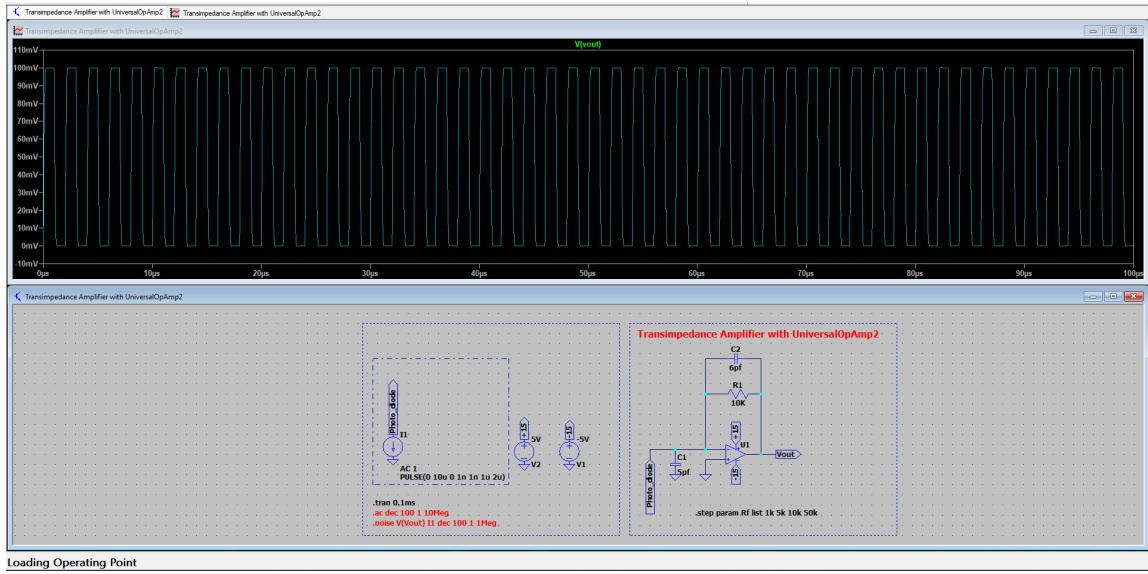


Figure 6: Step response for  $R_f = 10k$  and  $C_f = 6pF$  values (UniversalOpAmp2)

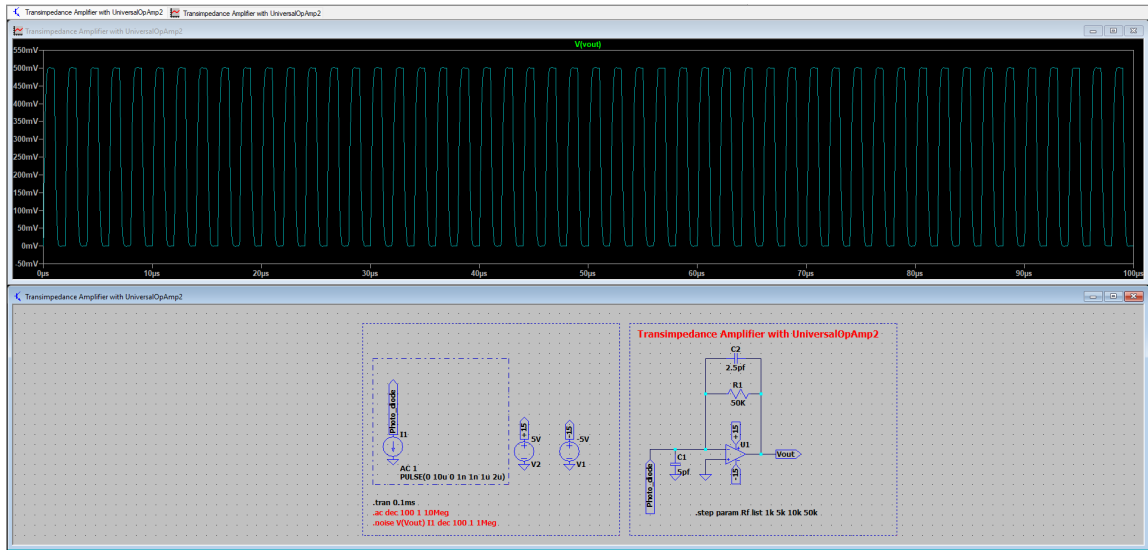


Figure 7: Step response for  $R_f = 50k$  and  $C_f = 2.5pF$  values (UniversalOpAmp2)

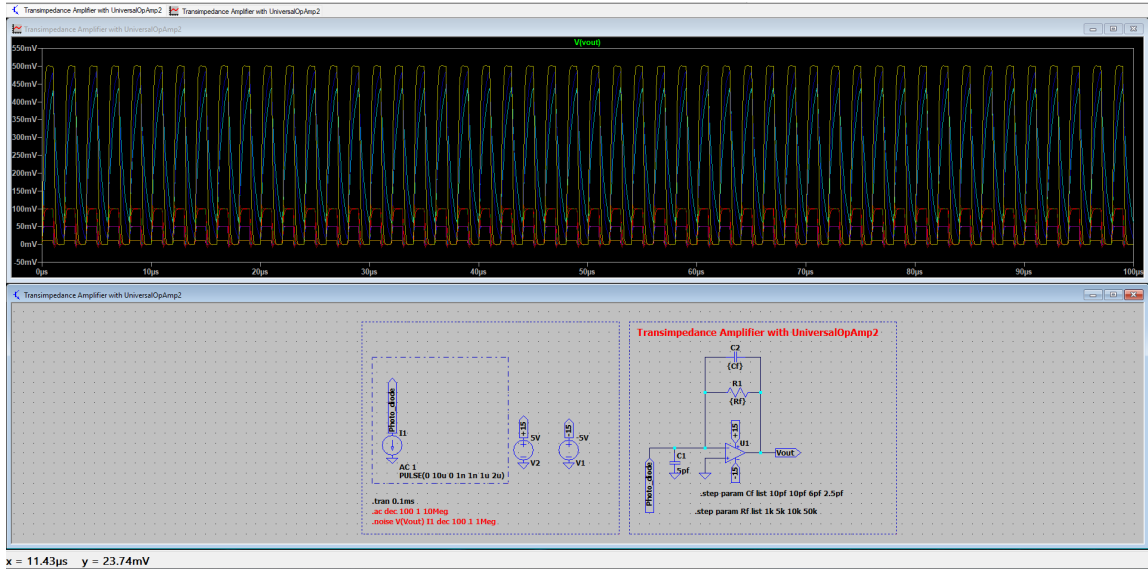


Figure 8: Step response for  $R_f = 1K, 5K, 10K, 50K$  and  $C_f = 10pF, 10pF, 6pF, 2.5pF$  values (UniversalOpAmp2)

## 5 TIA Using LTC6268

### 5.1 LTspice Schematic

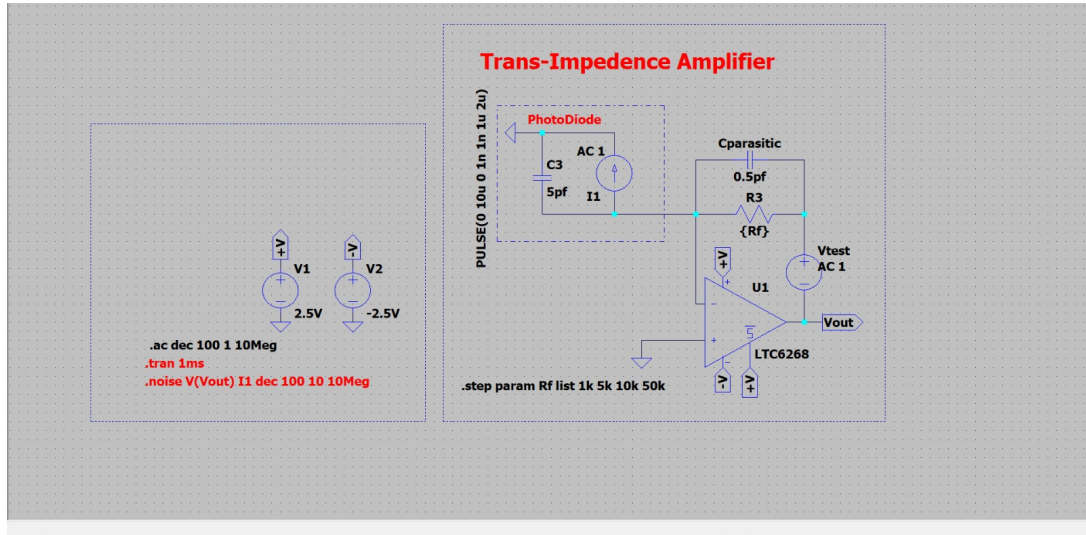


Figure 9: Transimpedance amplifier using LTC6268

## 5.2 Frequency Response and Stability

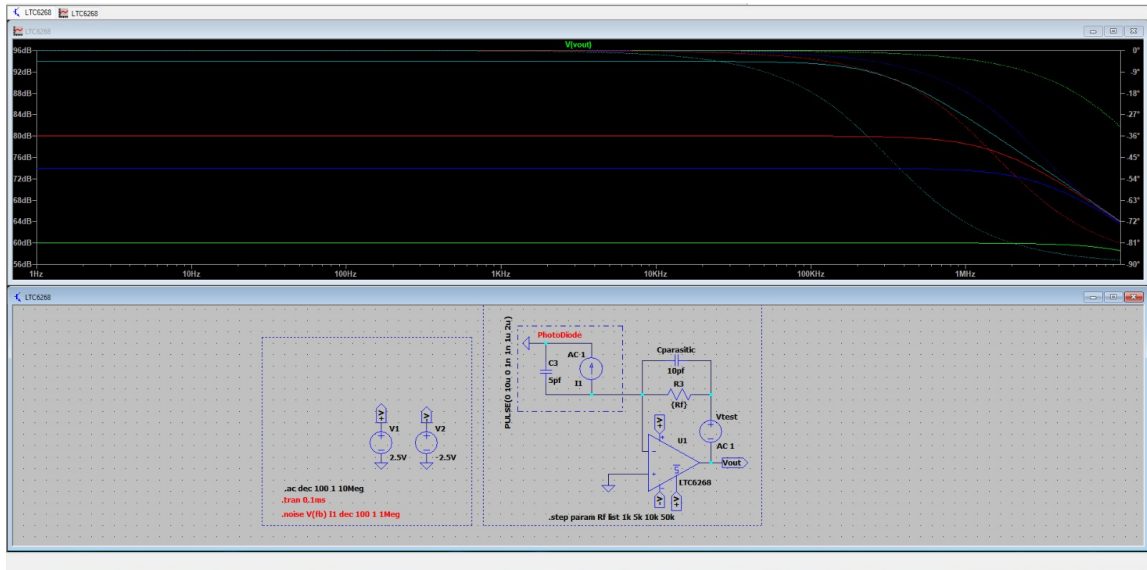


Figure 10: Bode plot (gain and phase) for LTC6268 TIA

The LTC6268 exhibits improved phase behaviour and higher usable bandwidth compared to the ideal op-amp model.

## 5.3 Noise Performance

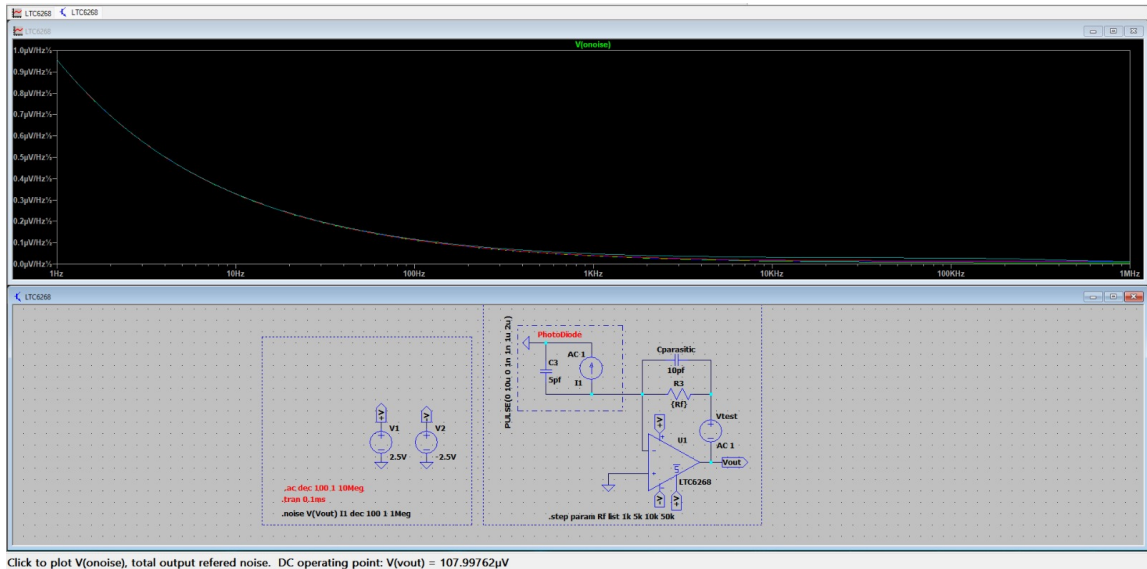


Figure 11: Noise spectrum for LTC6268 TIA

## 5.4 Transient Response

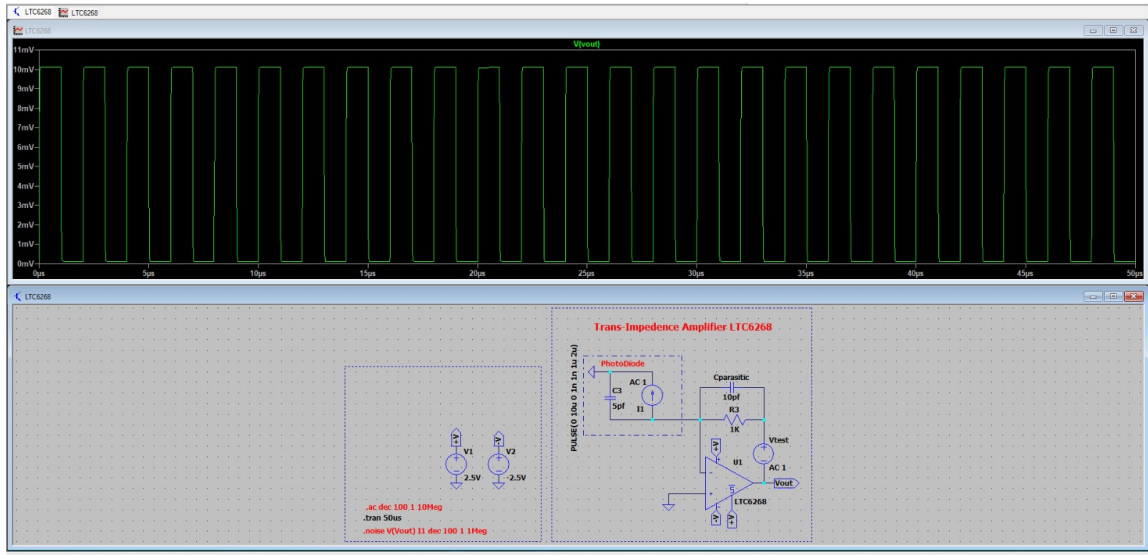


Figure 12: Step response for  $R_f = 1k$  and  $C_f = 10pF$  values (LTC6268)

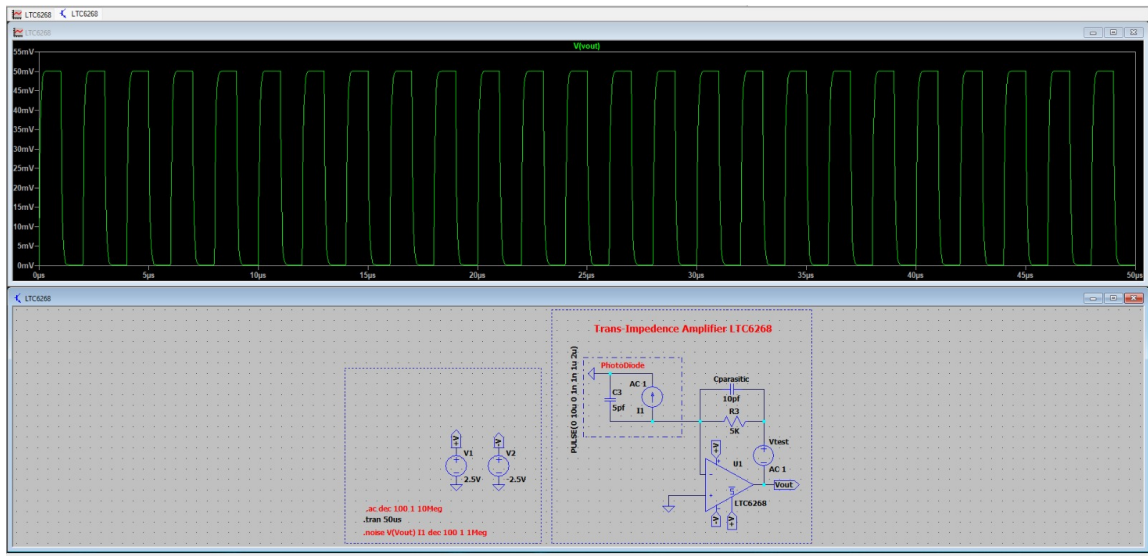


Figure 13: Step response for  $R_f = 5k$  and  $C_f = 10pF$  values (LTC6268)

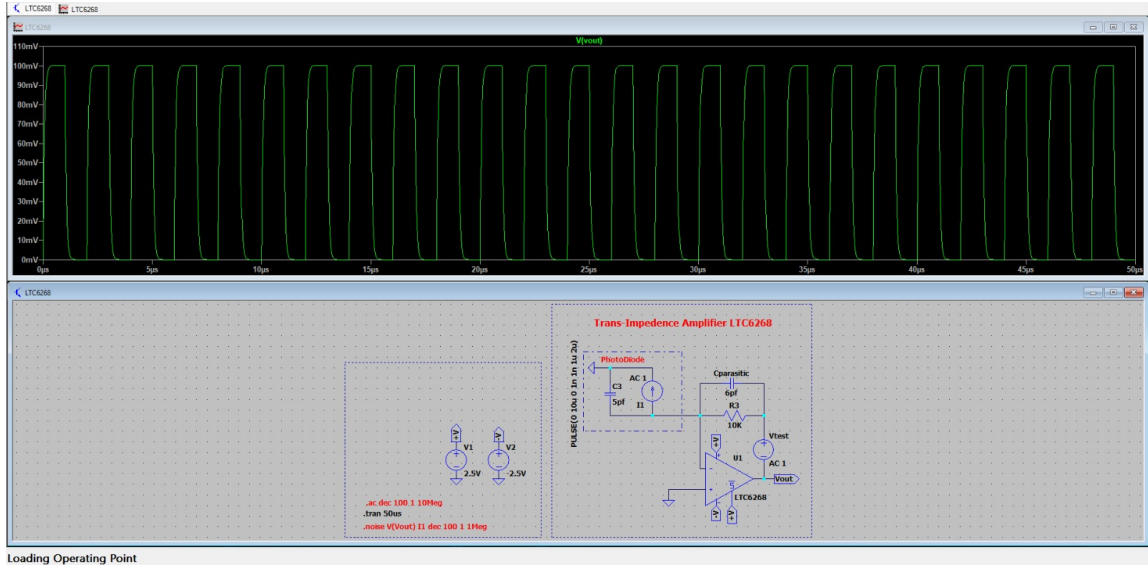


Figure 14: Step response for  $R_f = 10k$  and  $C_f = 6pF$  values (LTC6268)

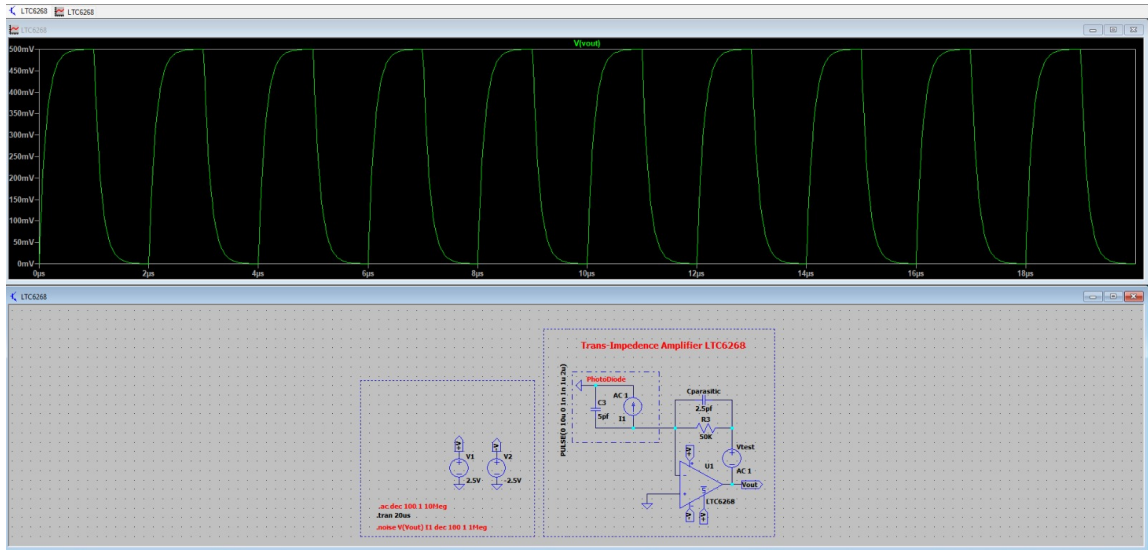


Figure 15: Step response for  $R_f = 50k$  and  $C_f = 2.5pF$  values (LTC6268)

## 6 Comparison and Discussion

- UniversalOpAmp2 is useful for conceptual understanding and trend analysis
- LTC6268 provides:
  - Better noise performance
  - Improved phase margin
  - More realistic TIA behaviour
- Increasing  $R_f$  increases gain but reduces bandwidth

- Proper choice of  $C_f$  is essential for stability

## 7 Conclusion

A complete transimpedance amplifier design was successfully simulated using LTspice. All stability, bandwidth, noise, and transient response requirements were verified for multiple feedback configurations. The comparison highlights the importance of using a TIA-optimised op-amp for precision photodiode applications.